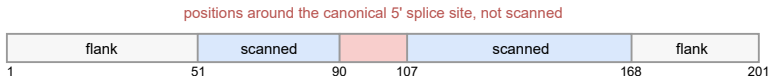


A The library: SMN2 exon 7, 2,000 cryptic 5'ss at 100 positions



2,000 cryptic 5' splice sites $\times$ 100 positions = 200,000 sequences

```
SCAN_POSITIONS = list(range(51,90)) + list(range(107,168))
```

B PoolParty

```
import poolparty as pp

# The 2,000 cryptic sites, as a pool
cryptic_pool = pp.from_seqs(
    cryptic_sites, mode="sequential")

# Replace one site at each position
library = pp.from_seq(target_region)\
    .replacement_scan(
        cryptic_pool,
        positions=SCAN_POSITIONS,
        mode="sequential")

# Final library, raw sequence strings
library_df = library.generate_library()
```

C tangermeme

```
import torch
from tangermeme.ersatz import substitute
from tangermeme.utils import one_hot_encode

# The 2,000 cryptic sites, as a tensor
cryptic_ohe = torch.stack(
    [one_hot_encode(site)
     for site in cryptic_sites])

# One copy of the region per cryptic site
target_ohe = (one_hot_encode(target_region)
               .unsqueeze(0)
               .expand(len(cryptic_sites), -1, -1))

# Substitute at each position,
# Final library, one-hot encoded tensor
library_ohe = torch.stack(
    [substitute(target_ohe, cryptic_ohe, start=p)
     for p in SCAN_POSITIONS])
```